Controlled STSP weight comparison

Shared population fits and ENF; $h_1 = 3\sigma$, $h_2 = 6\sigma$.

A. Full-grid log-log slopes of maximum error

Geometry	**Single**	**Naive** h^2	**Proposed**	**MS-STSP**
Circle	2.00142	1.99985	4.00412	1.94115
S2 in R3	2.00183	1.99989	4.00545	1.97880
Clifford torus	2.00264	1.99989	4.00781	1.96447
S3 in R4	2.00221	1.99996	4.00669	1.98686
S2 in R10	2.00167	1.99996	4.00500	2.07779
Corrugated torus	2.00675	1.99959	3.81725	2.40315

B. Unit-circle normalized errors and theoretical constant

σ	**Naive**/σ^2	$\Gamma_{1,2}$	**Rel. diff. (%)**	**Proposed**/σ^4
0.008	0.476369353	0.476376835	0.001571	1.747072
0.01	0.476365132	0.476376835	0.002457	1.747503
0.0125	0.476358521	0.476376835	0.003844	1.748176
0.016	0.476346745	0.476376835	0.006316	1.749371
0.02	0.476329630	0.476376835	0.009909	1.751099
0.025	0.476302612	0.476376835	0.015581	1.753806
0.03	0.476269133	0.476376835	0.022609	1.757127

All slopes use seven levels: 0.008, 0.01, 0.0125, 0.016, 0.02, 0.025 and 0.03. Finite-range slopes are measurements, not asymptotic proofs.

The proposed corrugated-torus slope is 3.81725 on the full grid and 3.91030 on the four smallest levels. The latter is only a sensitivity diagnostic.

Gamma(1,2) = 0.476376834958256965... is computed independently at 70-digit precision. Relative difference compares the measured normalized Naive error with this constant.

MS-STSP is a separate hard-ball population adaptation with its own iteration. Only Single, Naive and Proposed share the exact same two fitted points and correspondence.

Maximum errors use the frozen E1 query sets; the corrugated torus uses 128 fixed queries. No finite-sample Monte Carlo is used.